

Zhufeng (Zephyr) Qiu

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RESEARCH INTERESTS

GPU Systems | Compression-Communication Co-Design | HPC | ML Systems | Distributed LLM Training & Inference

RESEARCH SUMMARY

I am interested in building efficient and scalable systems for large-scale ML/AI workloads, with a focus on GPU runtime optimization, collective communication, and distributed execution. I have worked on parallel data processing, production-scale distributed systems, and performance engineering across serial CPU, OpenMP, MPI, CUDA, and NCCL backends, including a lossless homomorphic compression for NCCL AllReduce payloads and a controlled study of when communication-computation overlap helps. My Ph.D. interests are in HPC and ML systems, particularly communication-efficient execution and high-performance LLM training and inference.

EDUCATION

Northeastern University, Khoury College of Computer Sciences

Jan. 2023 - May 2025

- M.S. in Computer Science, GPA: 4.0/4.0
- Relevant coursework: Parallel Data Processing, Algorithms, Computer Networking, Cloud Computing.
- Teaching Assistant for Algorithms and Parallel Data Processing.

University of Southern California, Viterbi School of Engineering

Aug. 2019 - May 2021

- M.S. in Applied Data Science, GPA: 3.83/4.0
- Relevant coursework: Scientific Computing and Visualization, Analysis of Algorithms, Machine Learning, Data Mining, Database Systems.

Wuhan University, School of Resource and Environmental Sciences

Sep. 2014 - Jun. 2018

- B.S. in Geographical Information Science, GPA: 3.58/4.0
- Relevant coursework: Data Structures, Linear Algebra, Discrete Mathematics, Probability and Mathematical Statistics, C Programming.

Additional coursework: Operating Systems — Peking University via Coursera (non-credit) [Certificate](#)

Sep. 2026

SELECTED SYSTEMS, HPC & ML PROJECTS

BoundRelay — LLM KV-Cache Compression and Transfer Study [GitHub](#)

Aug. 2026 - Sep. 2026

Independent ML-systems study | Python, PyTorch, CUDA, cuSZp, SZ3, CUDA zfp

- Replicated K/V compression-sensitivity asymmetry in Qwen3-1.7B on **32 held-out articles**: at similar payload sizes, K-only compression raised mean teacher-forced NLL by **0.0155 (95% CI [+0.0068, +0.0235])**, while V-only compression did not.
- Built a benchmark covering **1,152 tensor observations**, with recorded model/codec revisions and binary hashes, frozen evaluation inputs and tolerances, and a versioned corrections register.
- Ran **360 paired prepared-cache comparisons**: approximately 3× payload compression cut fsync-acknowledged write time by **26–49%**, but increased pipelined host-staged GPU-to-GPU transfer time to **1.52–2.33× raw**. Host staging bypassed peer-copy failures that returned CUDA success without transferring data on all three rented multi-GPU pods.
- Identified per-tensor overhead and payload-dependent bandwidth as sources of prediction error; stage costs measured on the actual cache and hardware predicted **21.43 ms versus 43.65 ms** for the complete pipeline.

SGLang-Omni — Fun-ASR Context-Budget Fix [PR #2296 \(open\)](#)

Sep. 2026

- Submitted a fix for Fun-ASR's language-dependent context-budget error to restore **30-second English audio admission**; added regression tests for default, Chinese, and English prompts and validated on an RTX 4090.

Multi-GPU Similarity Engine - Lossless Homomorphic Collective Compression & Overlap [GitHub](#)

Jul. 2026 - Aug. 2026

Independent systems study extending USC INF 553 Data Mining / CSCI 596 Scientific Computing Projects | C++, CUDA, NCCL, MPI, OpenMP, PySpark, CMake, Nsight Systems

- Compressed the NCCL AllReduce payload **3× losslessly (56.2 → 18.8 MB)** by packing six integer sufficient statistics into two uint64 words with carry-free fields, letting NCCL **sum them in compressed form without decompressing**; verified bit-exact across 2.58M pairs.
- Isolated the interconnect by running the identical binary on NVLink and PCIe: communication is 11% vs 91% of the iteration, so the same compression reduces the iteration time by **4% on NVLink and 55% on PCIe**, and combined with overlap cuts the PCIe iteration **2.53×**.
- Re-engineered a Yelp collaborative-filtering workload from Spark into serial C++, OpenMP, MPI, CUDA, and dual-GPU NCCL backends, keeping all five bit-exact; processed **1.17M candidate pairs in 4.21 ms on two NVIDIA A100s, 312× faster than serial and ~20× faster than 16-thread OpenMP**.
- Derived the overlap ceiling **min(compute, comm)/total** and verified it with Nsight traces; measured the largest overlap benefit when compute and communication times were comparable, and found and fixed a double-buffering race through a chunk-size sweep.

Quantized LLM Inference Study on a LoRA-Fine-Tuned Llama 3.1 8B [GitHub](#)

Jul. 2025 - Sep. 2025; Jul. 2026 - Aug. 2026

Independent inference-systems study extending an LLM-engineering course fine-tune | Python, PyTorch, CUDA, PEFT/LoRA, bitsandbytes, Hugging Face Transformers, Modal

- Identified a **2.50× end-to-end regression in the deployed configuration**: a bf16 compute dtype carried over from Ampere training onto Turing, which has no bf16 tensor cores. Localized entirely to prefill (4.42×, MFU collapsing to 3.2%) and isolated by an unquantized control agreeing within 0.8%.
- Showed **4-bit quantization buys capacity, not speed**: NF4 decode runs 1.69× slower than fp16 while moving 3.9× fewer bytes, reaching 9.8% of peak bandwidth against fp16's 64.1% (dequantization-bound, not bandwidth-bound).
- Built a CUDA-event inference benchmark across four quantization configurations on **Turing and Ampere**, decomposing latency into prefill and decode against an analytical roofline; showed **97% of the tokens are prefill but account for only 39% of total time**, with decode costing 54× more per token.
- Fine-tuned Meta Llama 3.1 8B with QLoRA (rank 32 / α 64 on attention projections, **27M trainable parameters, 0.34% of base**) under completion-only supervision on a curated 400,000-item dataset, and served it 4-bit NF4 on an NVIDIA T4 via Modal.

Zephyr Manus - Agent Execution and Tool-Orchestration Platform [Live Demo](#)

Mar. 2026 - May 2026

Independent Project | FastAPI, Redis Streams, PostgreSQL, Docker, Next.js, MCP/A2A

- Built a Planner-ReAct agent runtime supporting task decomposition, external-tool execution, persistent state, and human-in-the-loop wait/resume workflows.
- Developed Docker-isolated execution for Browser, Shell, and File tools and an event-driven FastAPI backend using Redis Streams, PostgreSQL, SSE, and MCP/A2A.

Distributed Apriori Mining on Yelp Dataset [GitHub](#)

Feb. 2024 - May 2024

Northeastern CS 6240 Parallel Data Processing | Java, Hadoop MapReduce, HBase, Hive, AWS S3

- Built a distributed Apriori pipeline on Hadoop MapReduce over ~550K Yelp reviews and ~15K businesses, decomposing candidate generation, support counting, pruning, and iterative frequent-itemset discovery, with Hive external tables preprocessing raw JSON into per-user review baskets.
- Implemented two interchangeable storage backends for cross-pass state: an HBase path materializing each pass as a table with support filtering applied at write time, and a file-based path over HDFS/S3 with a runtime flag switching between local and S3 discovery of per-pass outputs.

Train Ticket Flash-Sale System [GitHub](#)

Jan. 2025 - Sep. 2025

Independent Project | Java, Spring Boot, Spring Cloud, Redis, RocketMQ, MySQL

- Built a Redis/RocketMQ-backed ticketing service with asynchronous order processing, idempotent requests, and inventory consistency safeguards against duplicate purchases and overselling.

RESEARCH EXPERIENCE

PM2.5 Inversion and Spatiotemporal Analysis for the Wuhan Metropolitan Area

Mar. 2016 - Oct. 2017

National Undergraduate Innovation and Entrepreneurship Training Program | Core Team Member | Advisor: Prof. Huanfeng Shen

- Integrated ground-monitoring measurements, NASA MODIS/AOD satellite observations, and meteorological data, performing preprocessing and spatiotemporal matching across multiple data sources.
- Contributed to the development and evaluation of a mixed-effects model for PM2.5 inversion, using cross-validation and baseline comparisons to assess predictive performance.
- Used MATLAB/R for modeling and ArcGIS to generate spatially continuous PM2.5 concentration maps across the Wuhan “1+8” metropolitan area; the national program passed final review with a **Good** rating.

Web-Based Dynamic Thematic Mapping Technologies

Mar. 2018 - May 2018

Undergraduate Thesis | SinoMaps Press Collaboration | Advisors: Prof. Haihong Zhu and Yuntao Long

- Surveyed web-based thematic-mapping methods and designed the overall, functional, and logical architecture for online visualization, interaction, and dynamic data updates.
- Built and validated a D3.js customs trade-map and map-service prototype supporting thematic visualization, statistical charts, interactive queries, and live data updates.

INDUSTRY EXPERIENCE

Beaconfire Solutions Inc. - Software Development Engineer

Jul. 2025 - Present

Seattle, WA (Remote)

- Developed enterprise HR and digital-commerce microservices with Spring Boot/Spring Cloud, integrating authenticated APIs, relational/document data stores, AWS S3, and RabbitMQ-backed asynchronous workflows.

NSFOCUS Technologies Group Co., Ltd. - R&D Engineer

Jun. 2021 - Dec. 2022

Wuhan, China

- Built and maintained distributed security-data pipelines using Kafka and Elasticsearch for multi-source log ingestion and normalization, and implemented Spark/MapReduce batch workflows serving 100+ enterprise clients and processing approximately 5M records per day.
- Developed host-anomaly detection features for illicit file transfers and backdoor processes using fuzzy inference, hierarchical clustering, and k-core graph mining over multidimensional security logs.
- Contributed to a production alerting workflow that achieved effective response for **100% of severe threats and 90% of high-risk threats** in a telecom deployment.

NSFOCUS Technologies Group Co., Ltd. - R&D Engineer Intern

Jul. 2018 - Mar. 2019

Wuhan, China

- Processed and normalized multi-source security logs covering user operations, file access, and process behavior to support downstream host-anomaly analysis.

TECHNICAL SKILLS

HPC / GPU: CUDA, NCCL, MPI, OpenMP, Nsight Systems, CMake

Languages: C/C++, Python, Java, Scala, SQL

ML / LLM: PyTorch, Transformers, PEFT/LoRA, bitsandbytes, scikit-learn, LangGraph, RAG, MCP/A2A

Systems / Data: Spark, Hadoop, Kafka, Elasticsearch, Redis, SQL databases, AWS, Docker, Spring Boot/Cloud, Git

HONORS

- Outstanding Student Leader, Wuhan University, 2014-2015 and 2015-2016.
- Outstanding Individual in Theoretical Study, Wuhan University, 2014-2015.